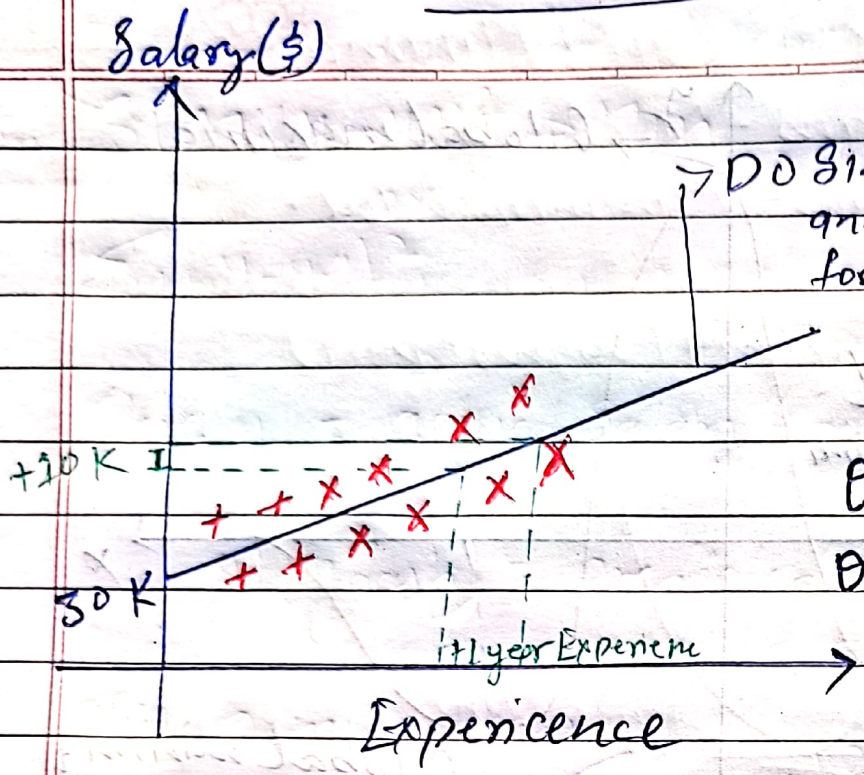
Linear Regression



→ Do simple linear regression and then he draw a line for to predict future value.

θ_0 = Intercept

θ_1 - Slope

Simple linear regression Equation = $h_\theta(x) = \theta_0 + \theta_1 \times x_1$

($h_\theta(x)$ = Output / Target Column)

- $h_\theta(x) = \theta_0 + \theta_1 \times x_1$ (x_1 = Input)

when something we have to predict we write on the $h_\theta(x)$.

- Salary = $\theta_0 + \theta_1 \times$ Experience

Salary = $\theta_0 + \theta_1 \times$ Exprience

Salary = we have to predict on the basis of Experience

θ_0 = Starting point (Fresher Salary is 30K) minimum Salary

θ_1 = If 1 year Experience then Increament of 10K

Experience - Year of Experience

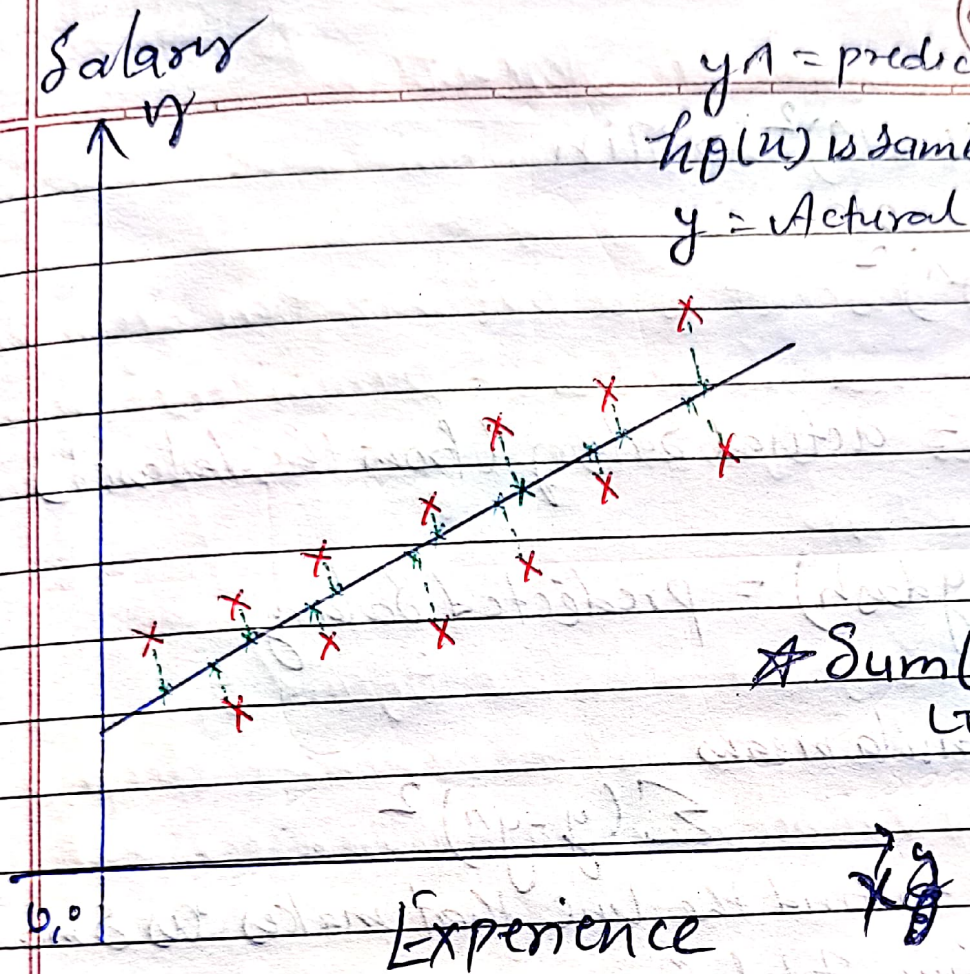
Ordinary Least Squares

Page No:

$y^\wedge$ = predicted Salary

$h_\theta(x)$ is same as $y^\wedge$ (y theta)

y = Actural Salary



$\star$ $Sum(Y-Y^\wedge)^2 \rightarrow min$
(Turnover)

Blue line (Regression line): It is the best fit-line Creted by the machine learning model after training

Green point (Predicted points) The green stars on the line represent the predicted salary values.

Green dash lines (Error): The vertical green dashed lines show the differente between real value and predicted value.

Redpoint (Actual points): This this real value from the dataset.

$\star$ The more close to redpoints more accuracy.

Ordinary Least Square

$SUM(y-y^{\wedge})^2 \rightarrow Min$

or

$\sum_i (y-y^{\wedge})^2$

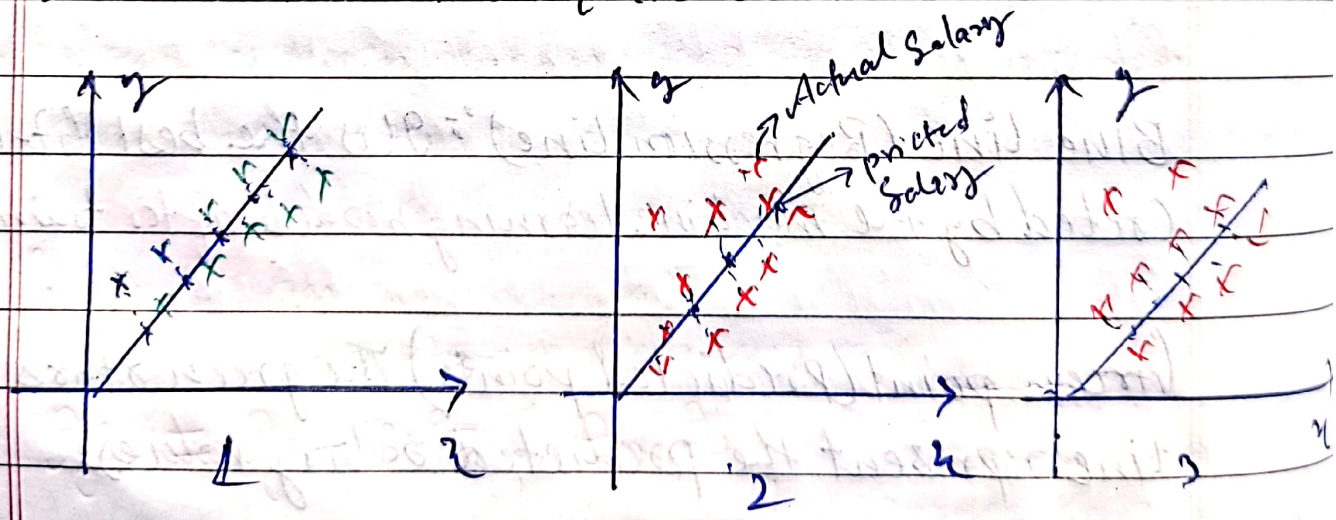
Here y = actual Salary (from the dataset)

$y^{\wedge}$ (ydash) = predicted Salary

This formula means

$$\sum_i (y-y^{\wedge})^2$$

we want to find the line that makes as small or minimum total square error



who has least or minimum total square error or who is closest to the linear line minimum error.

Turn over more clear

Cost Function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} [h_\theta(x^i) - y^{(i)}]^2$$

No need to Remember ML library already do things.

J = Cost (Error)

θ_0, θ_1 - ~~Has~~ Value has to minimise as possible the more minimum More accurary

m = Training set

~~h_θ - Which Target has to predict / output Column~~

~~predicted value~~

y = Actual value

$\frac{1}{2m}$ = So that value is not output very large.

$h_\theta(x)$ - predict value

$$\underbrace{J(\theta_0, \theta_1)}_{\text{Cost Fuction}} = \frac{1}{2m} \sum_{i=n}^{m} [\underbrace{h_\theta(x^i)}_{h_\theta(x) = \theta_0 + \theta_1 \times x_i} - y^{(i)}]^2$$

Main Goal is to Minimize the value of θ_0 and θ_1. is called

MEAN Squared Error Method